US 20250271000A1

(19) United States

(12) Patent Application Publication

(10) Pub. No.: US 2025/0271000 A1

(43) Pub. Date: Aug. 28, 2025

Woodruff et al.

(54) VENTILATION SYSTEM FAN SCROLL

(71) Applicant: **Broan-NuTone LLC**, Hartford, WI (US)

(72) Inventors: **Robert Woodruff**, Hartford, WI (US); **Jesse Hollingsworth**, Hartford, WI (US); **Anup Nikam**, Germantown, WI (US); **Luis Esparza**, Baja California (MX)

(21) Appl. No.: **18/585,704**

(22) Filed: **Feb. 23, 2024**

Publication Classification

(51) **Int. Cl.**

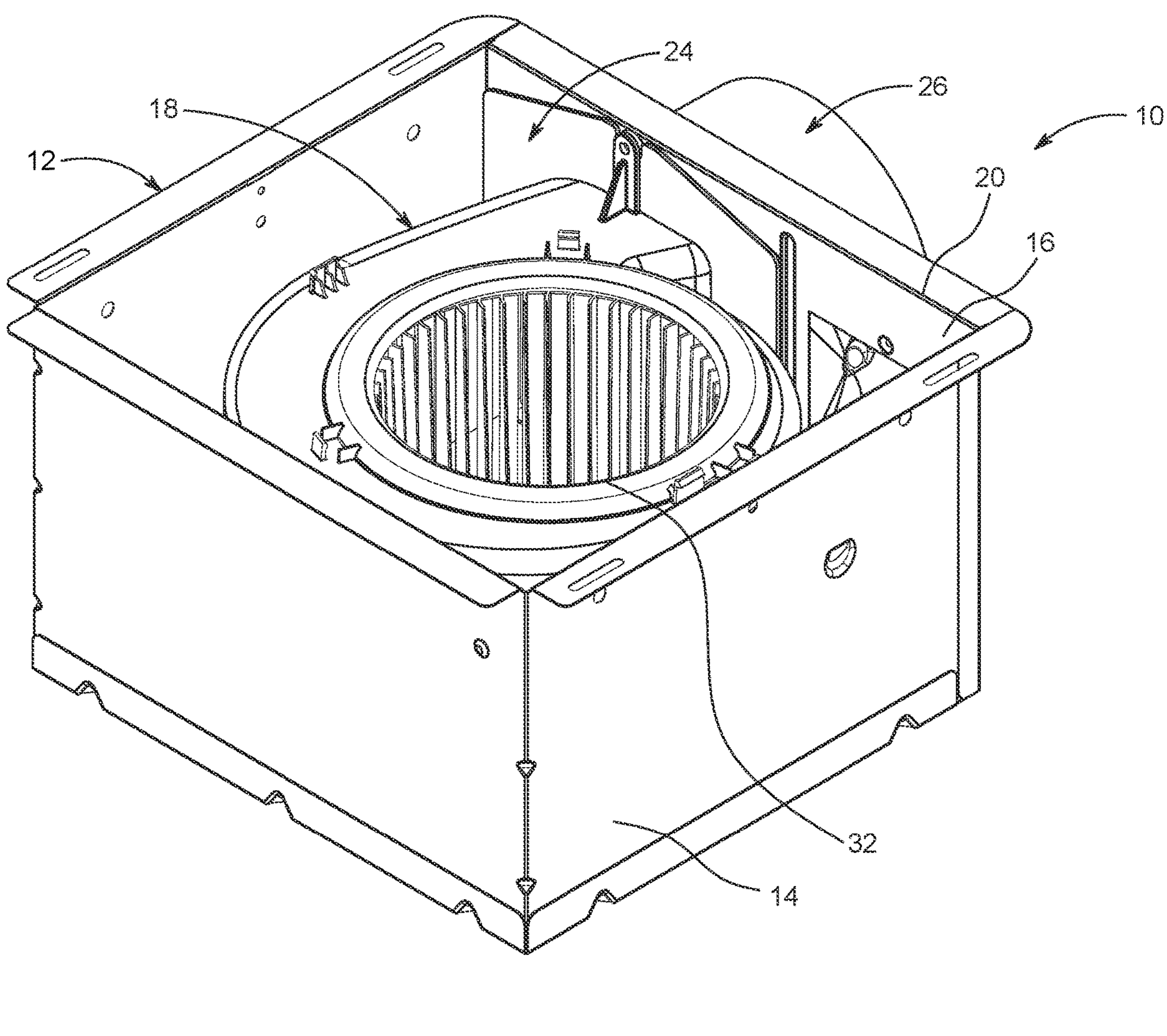
F04D 29/42	(2006.01)
F04D 25/08	(2006.01)
F24F 7/013	(2006.01)

(52) **U.S. Cl.**

CPC ***F04D 29/4226*** (2013.01); ***F04D 25/08*** (2013.01); ***F04D 29/422*** (2013.01); ***F24F 7/013*** (2013.01)

(57) **ABSTRACT**

A ventilation fan configured to be positioned adjacent to a room of a building structure to ventilate the room is disclosed. The ventilation fan has a fan housing with an external wall defining an internal region. A blower is configured to be located in the internal region. A tongue extends from one of the fan housing and the blower and a slot is defined by one of the fan housing and the blower. The slot configured to receive the tongue to secure the blower to the fan housing.



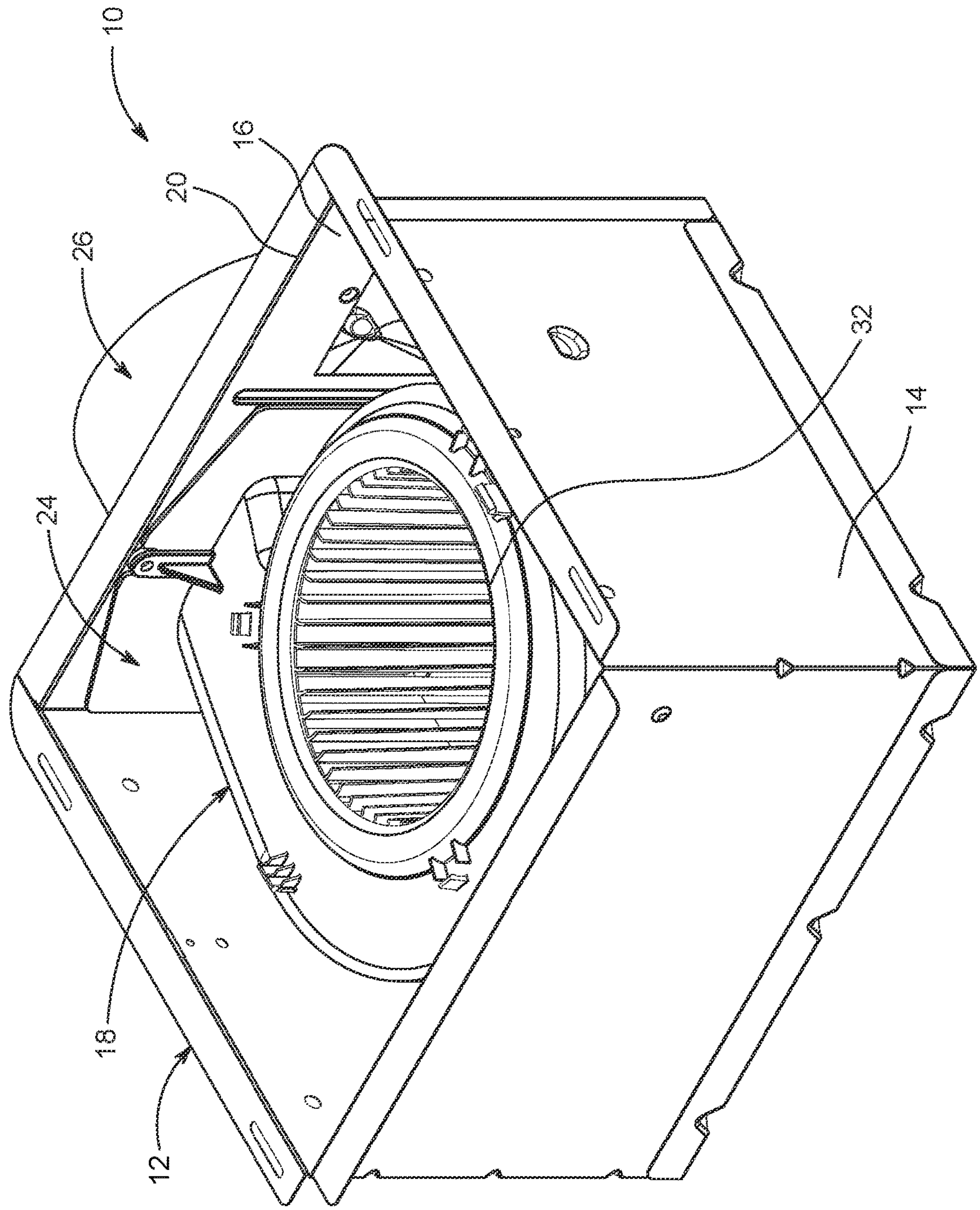


FIG. 1

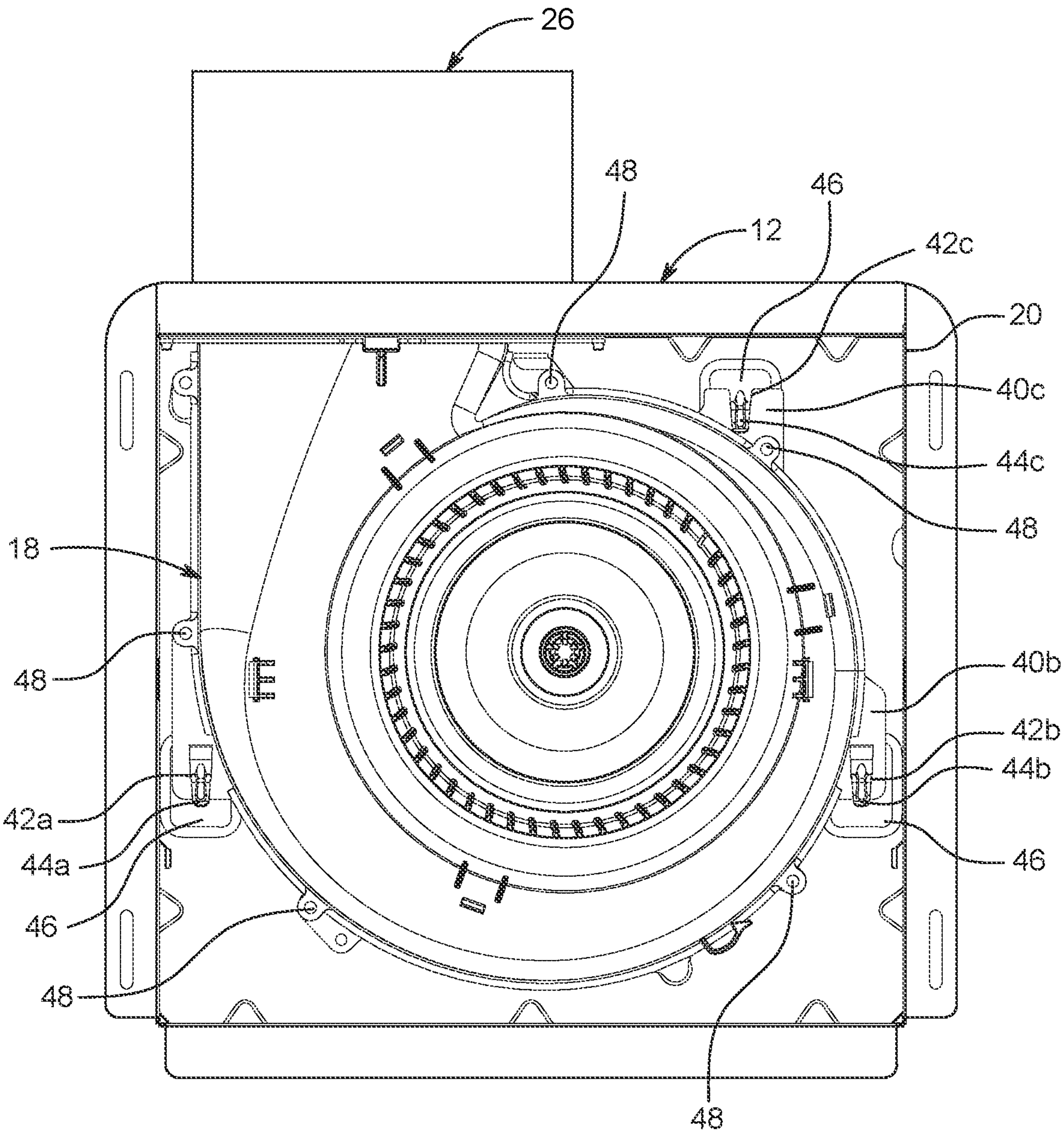


FIG. 2A

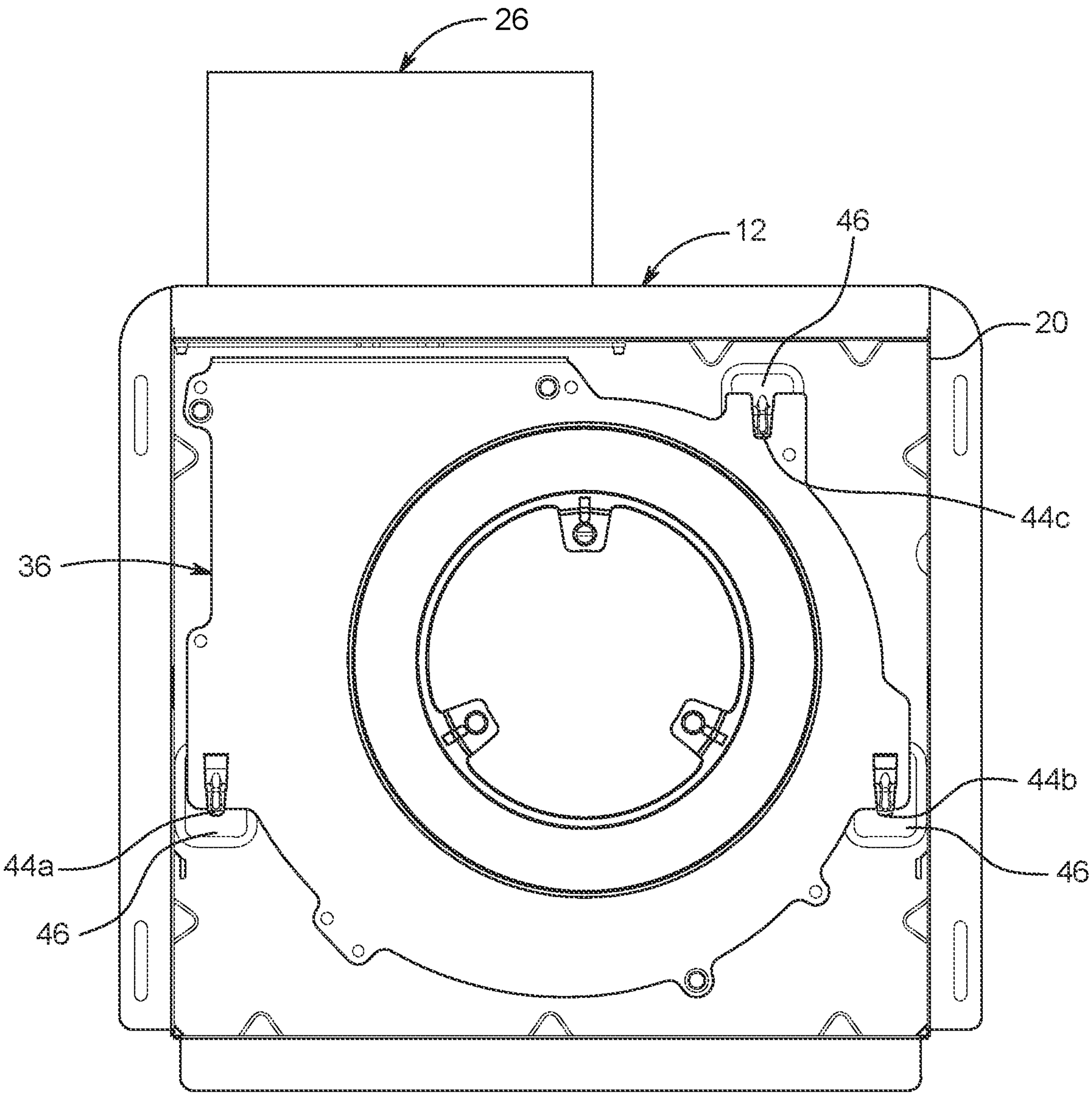


FIG. 2B

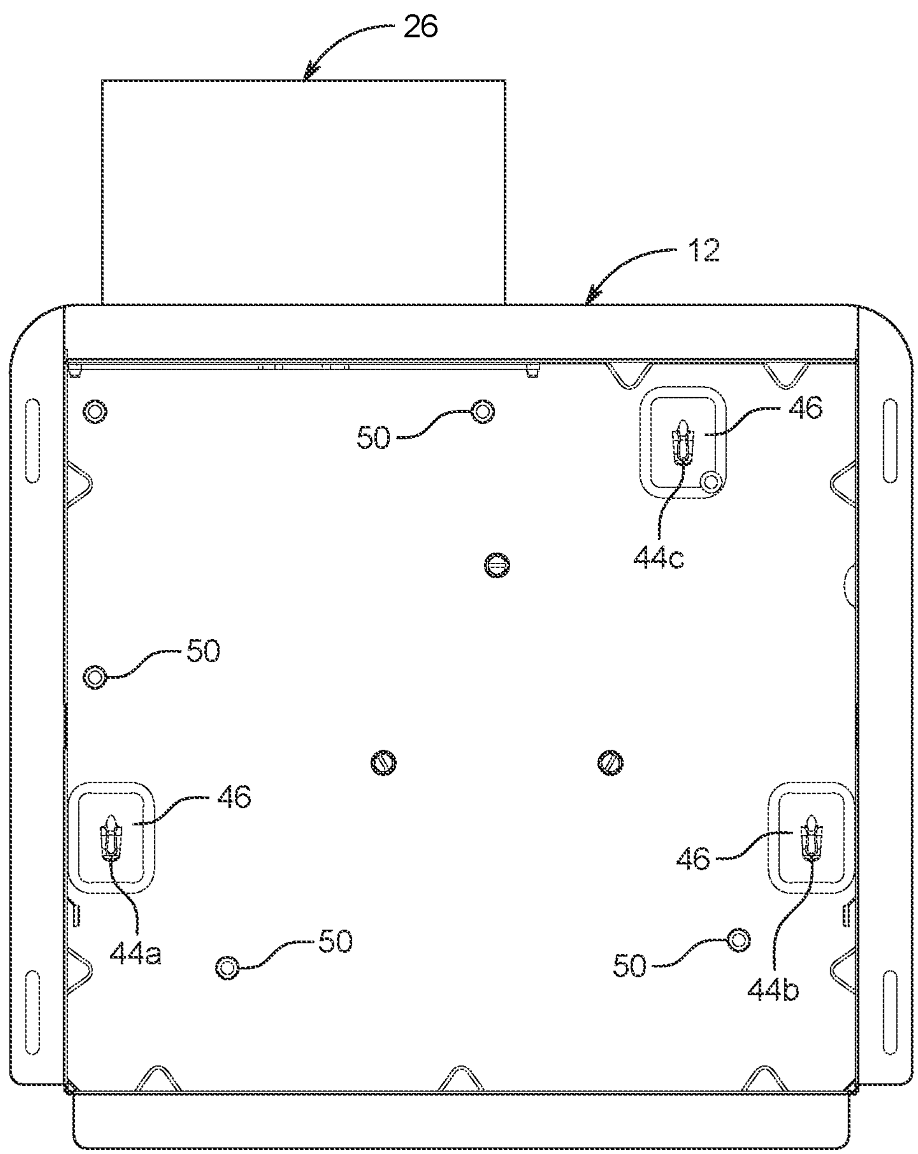


FIG. 2C

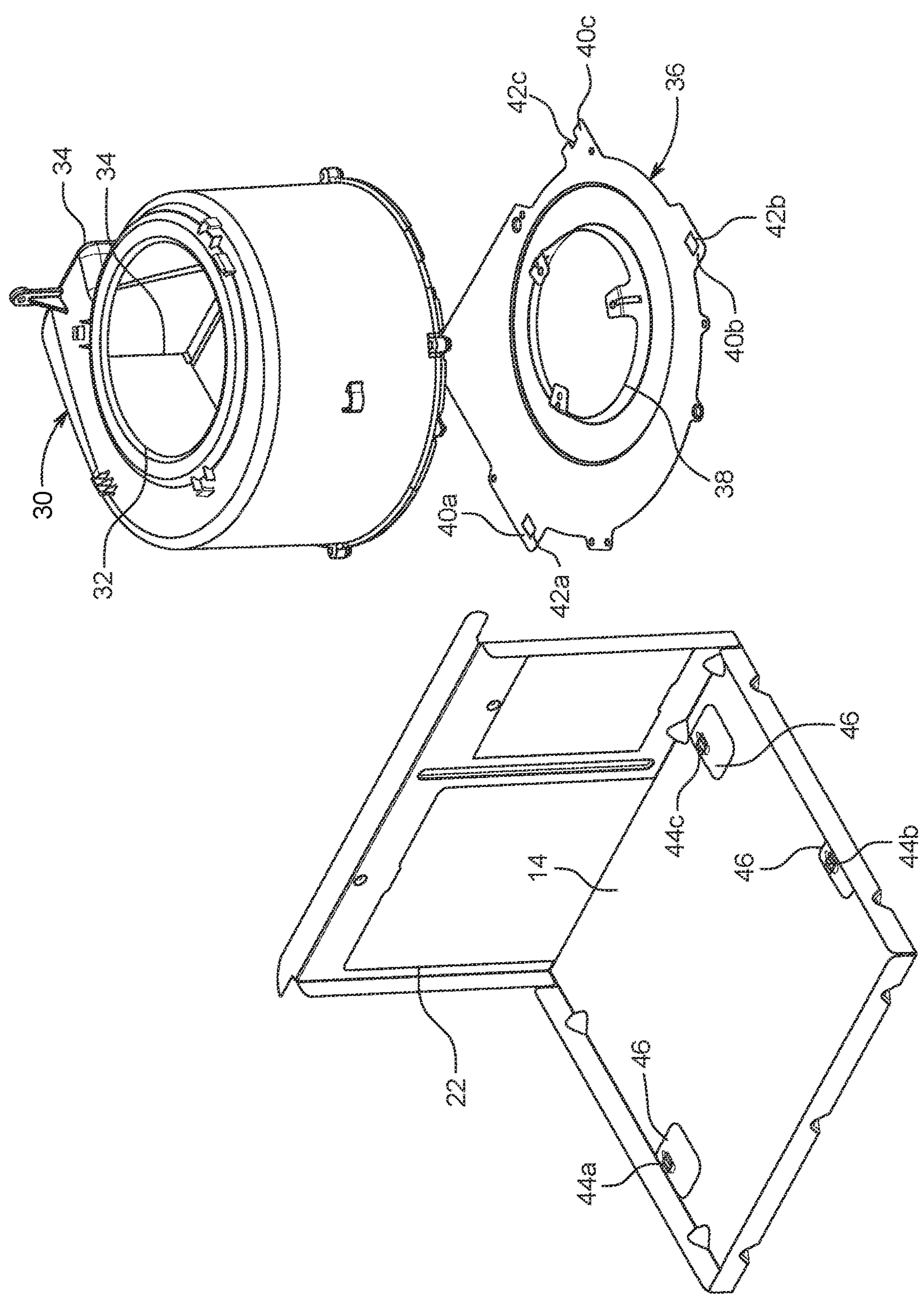


FIG. 2D

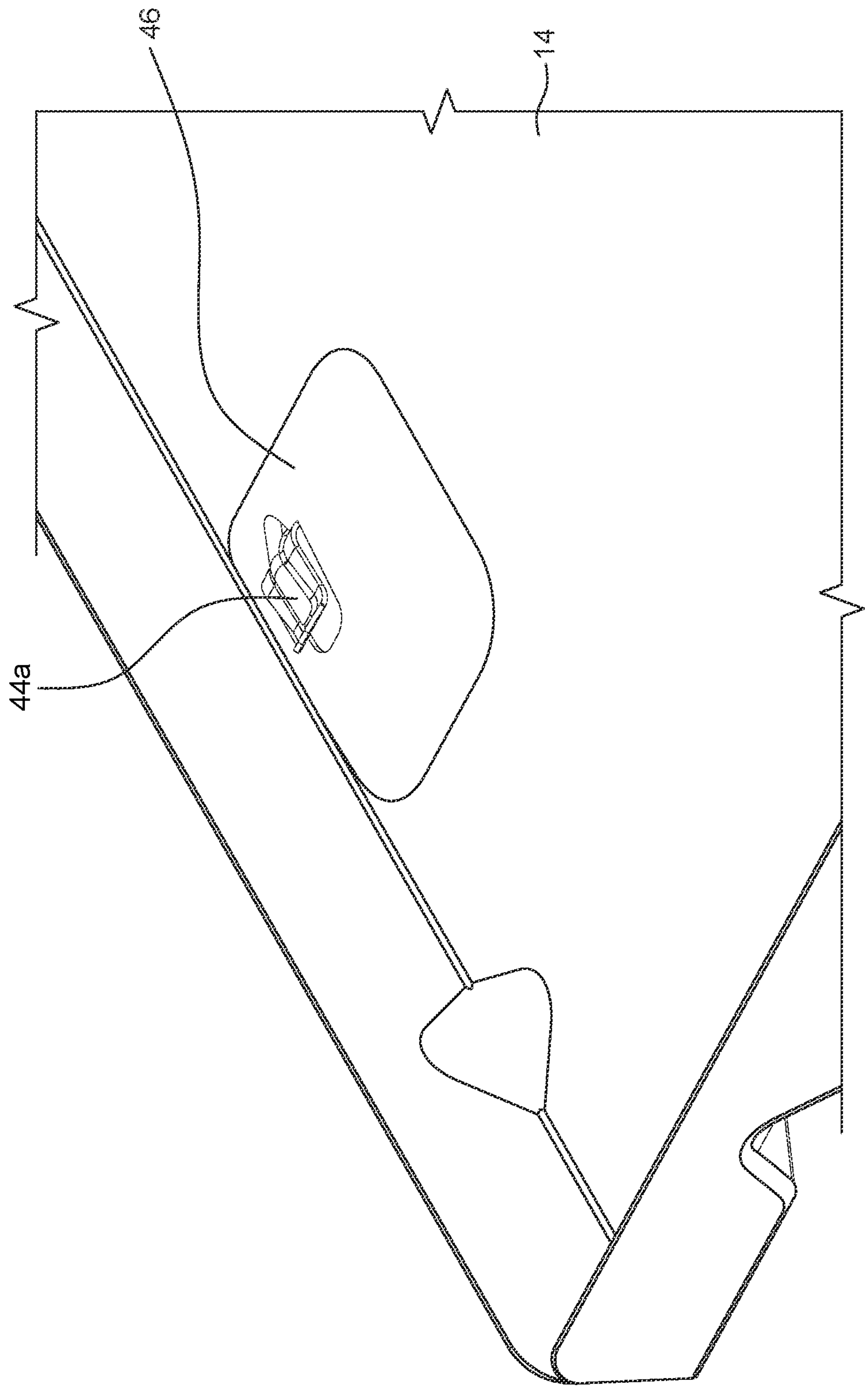


FIG. 3A

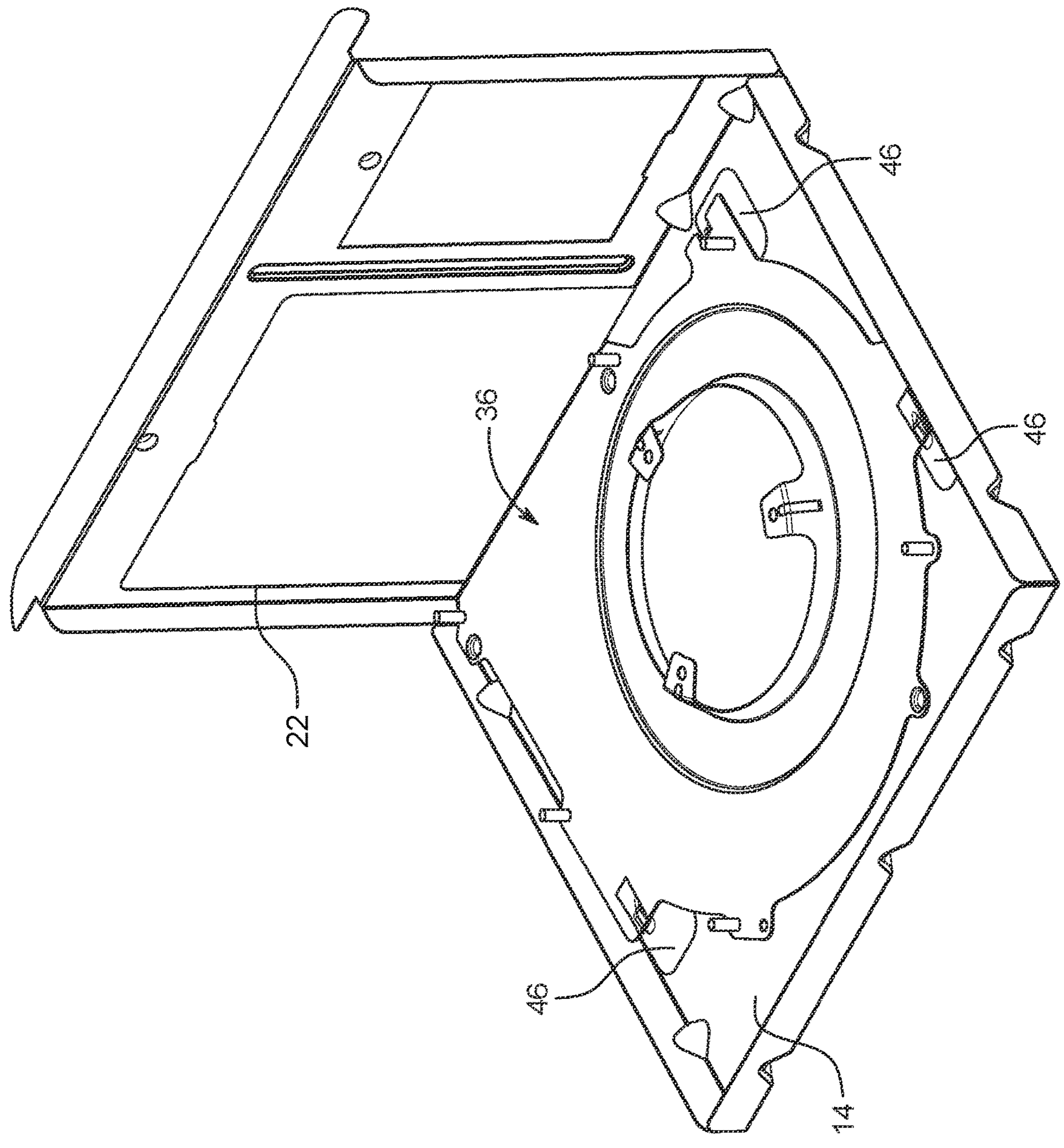


FIG. 3B

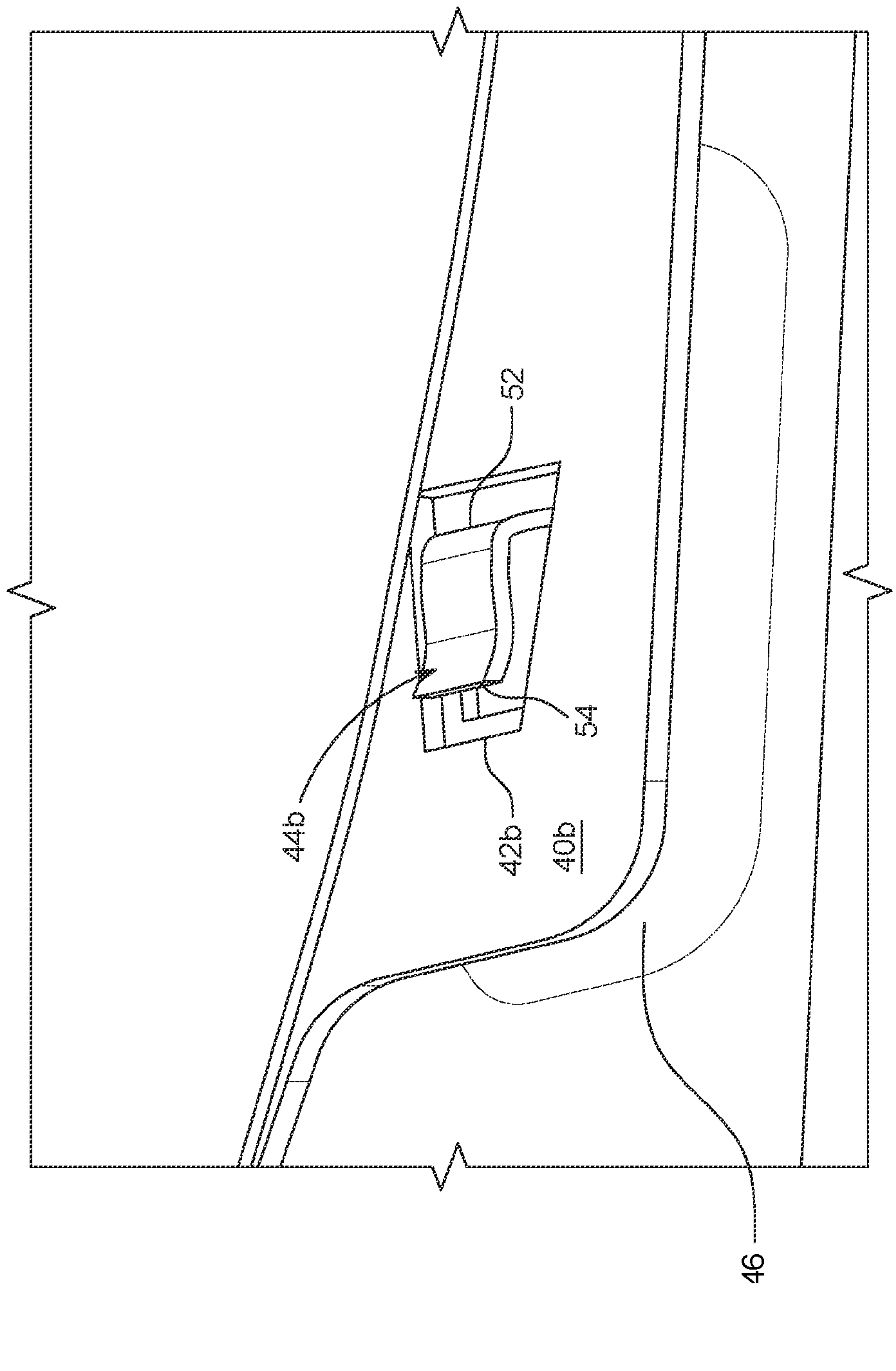


FIG. 3C

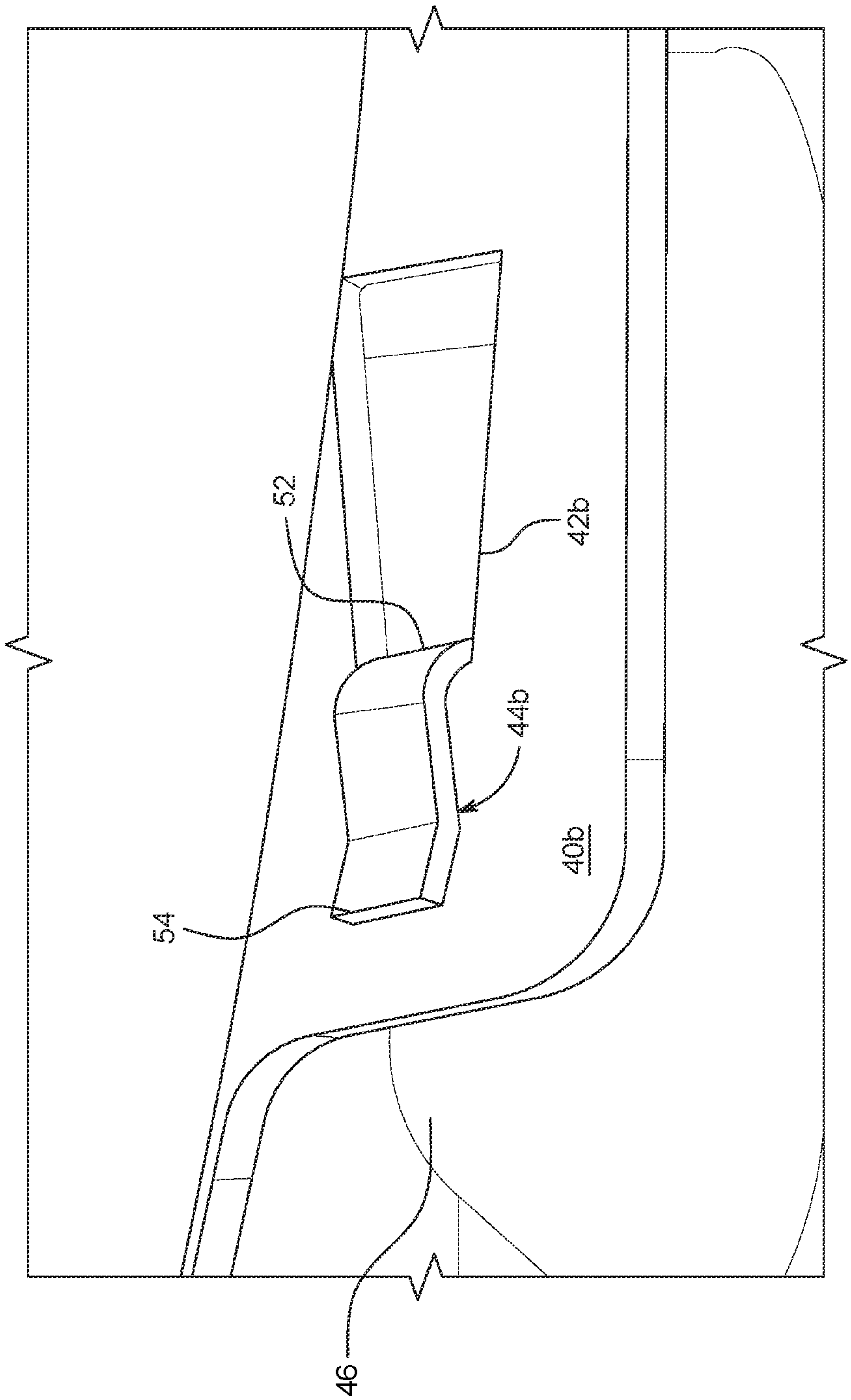


FIG. 3D

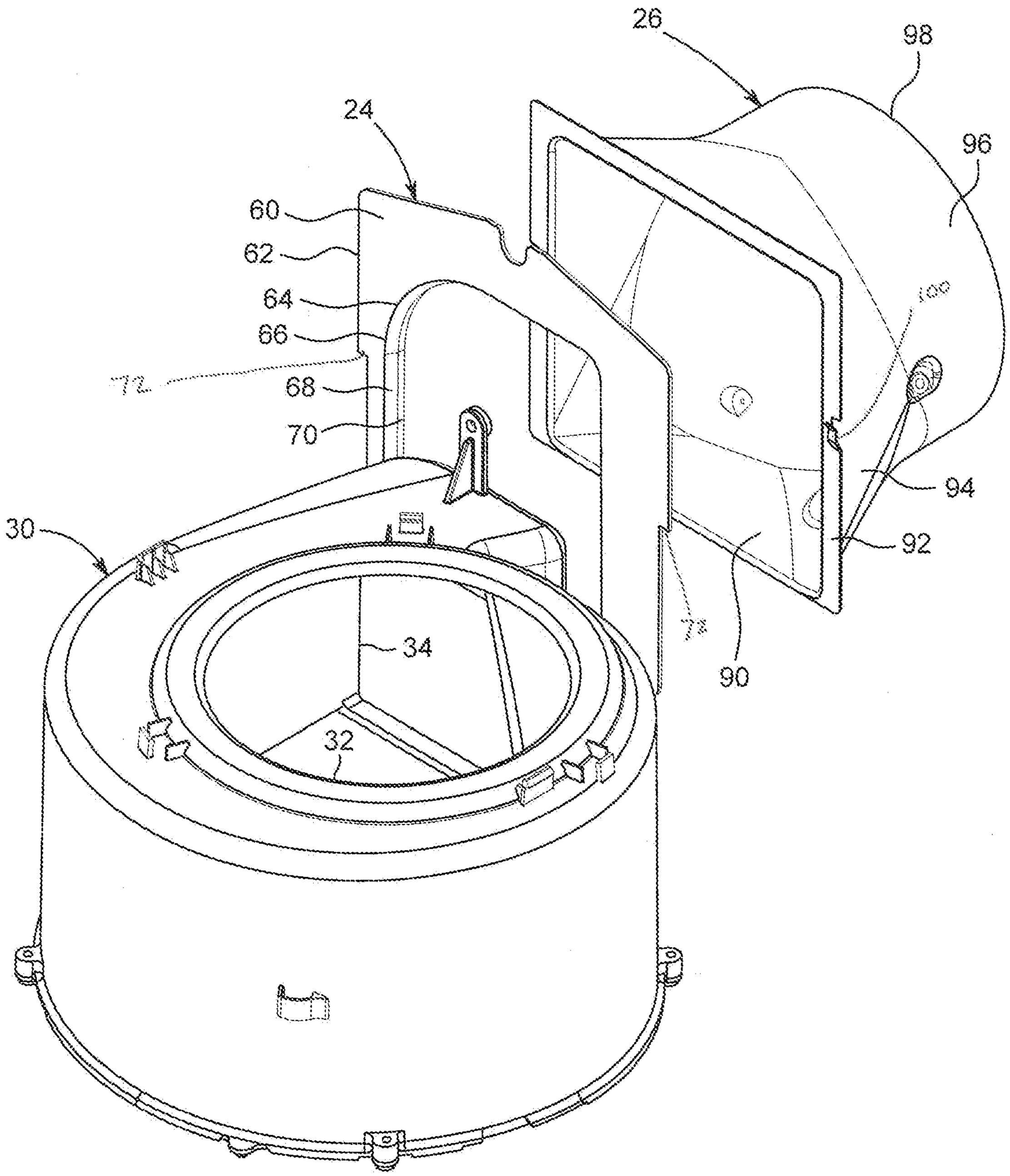


FIG. 4A

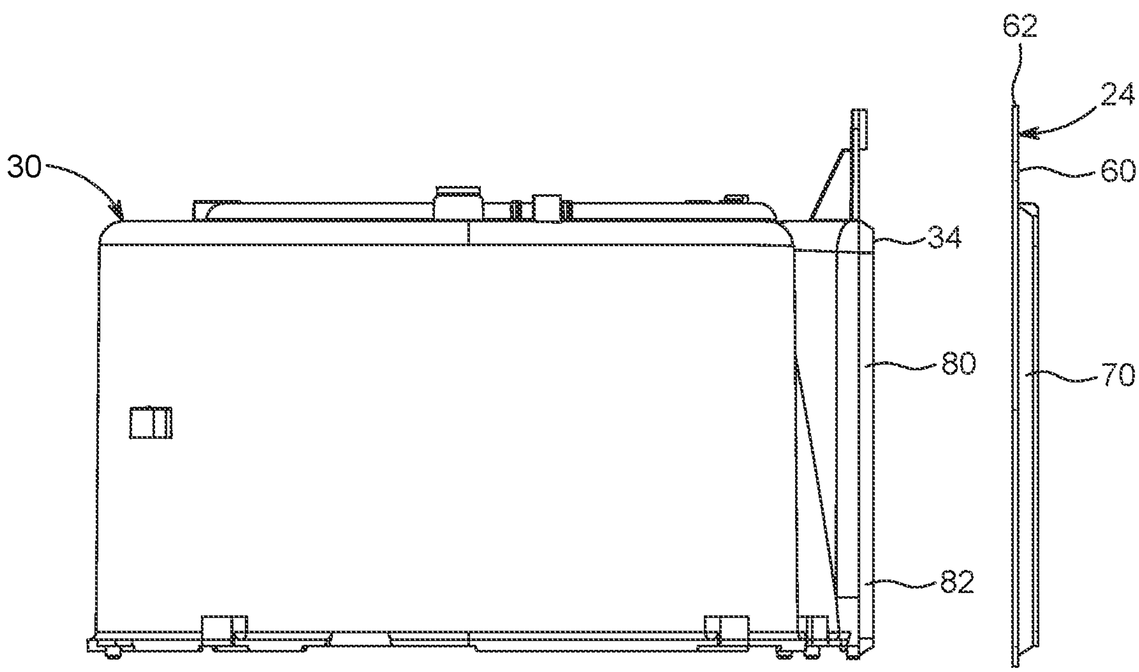


FIG. 4B

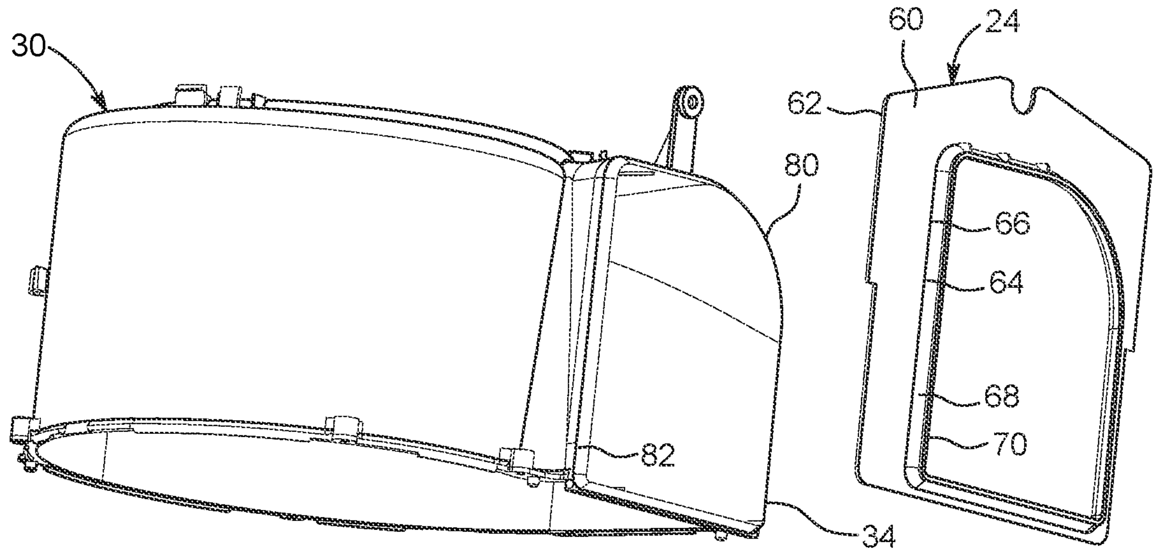


FIG. 4C

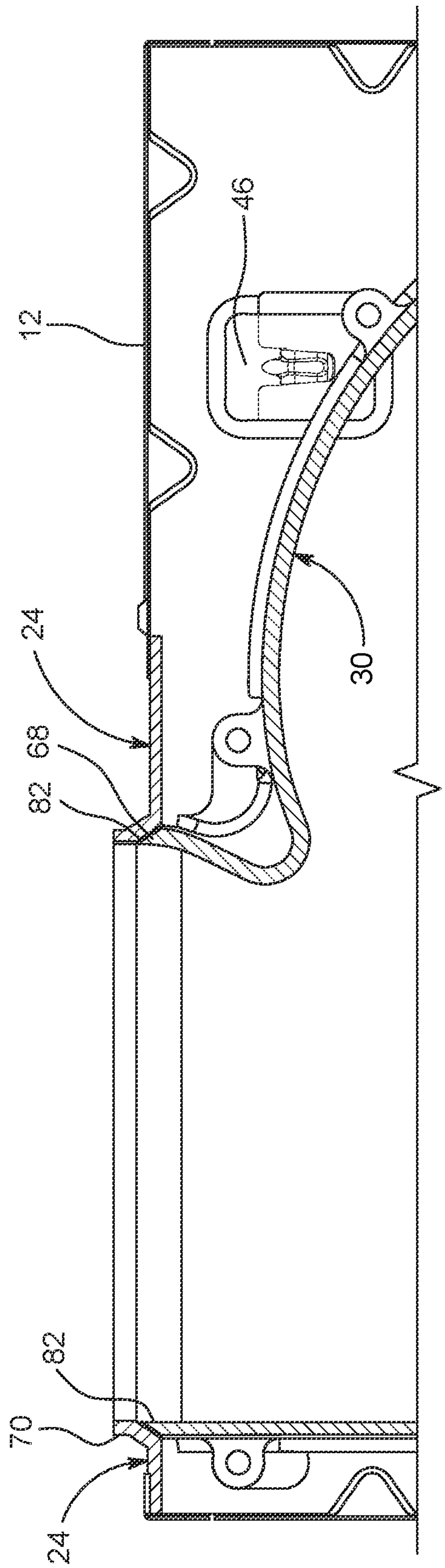


FIG. 4D

VENTILATION SYSTEM FAN SCROLL

BACKGROUND

[0001] Conventional ventilation fans, such as those typically installed in a room of a building structure (e.g. a bathroom) can draw air from within an area of the room, through the fan and exhaust the air to another location, such as through a vent in the gable or roof of a home or other building structure. Many conventional ventilation fans include a fan housing positioned within or adjacent an aperture formed in a wall or ceiling of the room and a blower secured to the fan housing by inserting it into the housing in one motion and securing the blower to the fan housing by fasteners. Keeping the fastener holes of the blower aligned with those of the fan housing during installation of the fasteners can be challenging. Therefore, a need exists for a ventilation fan that allows for easier securing of the blower to the fan housing and/or use of fewer fasteners to simplify assembly and reduce assembly time.

[0002] Conventional ventilation fans, such as the one described above, also use a duct connector that has a rectangular input (to match a rectangular opening in the fan housing) and a round output to facilitate connection with a typical round ducting used to direct outflow to the gable, roof or other portion of the building in which air is to be exhausted. Blowers are typically mounted in the fan housing with the blower outlet located adjacent to, and often pressed against, the fan housing outlet aperture. Airflow leakage often occurs at the interface between the blower and the fan housing. Similarly, the duct connector is typically mounted to the exterior of the fan housing adjacent the fan housing outlet aperture and likewise experiences airflow leakage. Therefore, a need exists for a ventilation fan that facilitates a better interface between the blower, the fan housing and the duct connector to reduce airflow leakage.

[0003] A full discussion of the features and advantages of the present disclosure is deferred to the following detailed description, which proceeds with reference to the accompanying drawings.

[0004] The description provided in the background section should not be assumed to be prior art merely because it is mentioned in or associated with the background section.

SUMMARY

[0005] A ventilation fan configured to be positioned adjacent to a room of a building structure to ventilate the room is disclosed, the ventilation fan has a fan housing with an external wall defining an internal region; a blower configured to be located in the internal region; a tongue extending from one of the fan housing and the blower; and a slot is defined by one of the fan housing and the blower, the slot configured to receive the tongue to secure the blower to the fan housing. The blower can have a scroll and a scroll mounting plate, and the slot can be defined in the scroll mounting plate. The ventilation fan can have a second tongue and a third tongue. The ventilation fan can define a second slot and a third slot. The tongue can have a proximal end at the fan housing and a distal end spaced from the fan housing, and the scroll mounting plate can sit between the tongue distal end and the external wall when the blower is secured to the fan housing. The scroll mounting plate can have an car extending beyond the perimeter of the scroll and the slot can define the car. The blower can be secured to the fan housing by sliding the scroll mounting plate under the tongue.

[0006] A ventilation fan is configured to be positioned adjacent to a room of a building structure to ventilate the room and the ventilation fan has a fan housing comprising an external wall defining an internal region, a fan housing inlet to the internal region and a fan housing outlet from the internal region; a blower configured to be located in the internal region, the blower having a scroll defining a blower inlet through which the blower is configured to draw air and a blower outlet through which the blower is configured to expel air, the blower outlet defining a blower outlet bevel; and an outlet seat between the blower outlet and the fan housing outlet, the outlet seat defining an aperture complementing the shape of the blower outlet and defining an outlet seat bevel; wherein the blower outlet bevel is configured to engage the outlet seat bevel. The outlet seat bevel can complement the blower outlet bevel. The ventilation fan can also have a duct connector that defines the outlet seat. The outlet seat can have a sleeve configured to extend into the fan housing outlet and the outlet seat bevel can extends into the sleeve. The outlet seat can have a plate defining the outlet seat aperture and a sleeve extending from the plate at the outlet seat aperture, wherein the sleeve comprises at least part of the outlet seat bevel, and the outlet seat bevel complements the blower outlet bevel. The plate of the outlet seat can be secured to the fan housing. A duct connector can be secured to the exterior of the fan housing, wherein the outlet seat has a sleeve configured to extend into the fan housing outlet and contact the duct connector outside of the fan housing.

[0007] A method for assembling a ventilation fan having a fan housing defining an internal region, and a blower, the method comprising the steps of: (a) moving the blower in a first direction toward the fan housing to place the blower into the internal region; (b) moving the blower a second direction, generally perpendicular to the first direction to secure the blower to the housing. The fan housing can comprise a tongue and the blower can comprise a slot and step (a) can further include locating the slot around the tongue. The tongue can extend from the fan housing into the internal region and step (a) can further include moving the blower in the first direction until the blower contacts the fan housing. Step (b) can further include moving the blower in the second direction until the blower is under the tongue. The blower can have an ear in which the slot is defined and step (b) can further include moving the blower in the second direction until the ear is between the tongue and the fan housing. The tongue can extend from the fan housing at a tongue proximal end and extend to a tongue distal end and step (b) can include moving the blower in the second direction until the ear contacts the tongue proximal end.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The accompanying drawings, which are included to provide further understanding and are incorporated in and constitute a part of this specification, illustrate disclosed embodiments and together with the description serve to explain the principles of the disclosed embodiments. In the drawings:

[0009] FIG. **1** is a perspective view of a ventilation system of the present disclosure.

[0010] FIG. **2**A is a top view of the ventilation system of FIG. **1**.

[0011] FIG. **2**B is the view of FIG. **2**A with the scroll of the blower removed to show the scroll mounting plate.

[0012] FIG. **2**C is the view of FIG. **2**A with the blower, including the scroll mounting plate, removed.

[0013] FIG. **2**D is an exploded view of the scroll, the scroll mounting plate and a portion of the fan housing of FIG. **1**.

[0014] FIG. **3**A is a perspective view of a portion of the fan housing of FIG. **2**D.

[0015] FIG. **3**B is a perspective view of the scroll mounting plate secured to a portion of the fan housing.

[0016] FIG. **3**C is a perspective view of the scroll mounting plate located over a portion of the fan housing in a first position;

[0017] FIG. **3**D is a perspective view of the scroll mounting plate located over a portion of the fan housing in a second position;

[0018] FIG. **4**A is a perspective, exploded view of the scroll, the blower outlet seat and the duct connector of the ventilation system of FIG. **1**.

[0019] FIG. **4**B is a side, exploded view of the scroll and the blower outlet seat.

[0020] FIG. **4**C is a perspective, exploded view of the scroll and the blower outlet seat.

[0021] FIG. **4**D is a top cross-sectional view of the interface between the scroll, the scroll interface and the fan housing.

[0022] In one or more implementations, not all of the depicted components in each figure may be required, and one or more implementations may include additional components not shown in a figure. Variations in the arrangement and type of the components may be made without departing from the scope of the subject disclosure. Additional components, different components, or fewer components may be utilized within the scope of the subject disclosure.

DETAILED DESCRIPTION

[0023] FIG. **1** depicts one embodiment of a ventilation fan **10** in accordance with the present disclosure. The ventilation fan **10** is configured to be positioned in a room of a building structure (not depicted) to provide ventilation for the room of the building structure. For example, the ventilation fan **10** may be positioned in a ceiling panel (not shown) of the room and aligned with a cutout in the ceiling panel to selectively ventilate the room. The ventilation fan **10** includes a fan housing **12** with an external wall structure **14** that defines an internal region **16**. A blower **18** is located in the internal region **16** of the fan housing **12**. The external wall structure defines an inlet **20** through which air can be drawn and an outlet **22** through which air can be expelled. A blower outlet seat **24** is located at the fan housing outlet **22** between the blower **18** and the fan housing **12**. The fan housing inlet **20** is cooperatively shaped and dimensioned to align with the cutout formed in the ceiling of the room of the building structure. A duct connector **26** is located on the exterior of the fan housing **12** at the fan housing outlet **22** to receive exhaust air from the blower **18** and direct that exhaust air to exhaust ducting (not shown).

[0024] The fan housing **12** can be formed of any material known to those skilled in the art capable of withstanding varying temperatures, namely to withstand any heat radiated and/or conducted from the blower **18** and/or other components while providing structural integrity to the ventilation fan **10**. In some embodiments, the fan housing **12** is formed of sheet metal, but could instead be formed of a ceramic or a polymer having a relatively high melting temperature and/or glass transition temperature. The fan housing **12** can have any shape, including a box-like or cubical shape, a hemi-spherical shape, a spherical shape, a pyramidal shape, and the like. The fan housing **12** can form a base or frame for the ventilation fan **10**, thereby providing points and areas of attachment for other components of the ventilation fan **10**.

[0025] In the depicted embodiment, the blower **18** is a centrifugal fan including a motor (not depicted) and a bladed rotor **28** that is configured to rotate about an axis and move air from the fan housing inlet **20** to the fan housing outlet **22** to provide ventilation to the room. However, other types of blower assemblies can be employed as desired. The blower **18** also includes a scroll **30** that partially encases the bladed rotor **28** and defines a blower inlet **32**, through which the blower **18** draws air, and a blower outlet **34** through which the blower **18** exhausts air. A scroll mounting plate **36** extends across the bottom of the scroll **30** to substantially close the bottom of the scroll **30** except for a motor aperture **38** to accommodate portions of the blower motor.

[0026] The depicted scroll mounting plate **36** has three ears **40***a*, **40***b* and **40***c* (collectively referenced as ears **40**) that, as best depicted in FIG. **2**A, each extend outward beyond the perimeter of the scroll **30**. A greater or lesser number of ears **40** could be used in alternative embodiments. Each of the ears **40** defines a slot **42***a*, **42***b* and **42***c* (collectively referenced as slots **42**). In the depicted embodiment, two of the slots **42***a*, **42***b* define apertures, while the third slot **42***c* defines a notch. Depending on the location of the ears in the fan housing **12**, some, all or none of the slots **42***c* could be notches instead of apertures. In an alternative embodiment, the ears could be defined by the scroll **30** instead of the scroll mounting plate **36**.

[0027] The bottom wall of the external wall structure **14** of the fan housing **12** comprises three tongues **44***a*, **44***b* and **44***c* (collectively referenced as tongues **44**) extending from the fan housing **12** and protruding into the internal region **16** of the fan housing **12**. The depicted embodiment includes optional pedestals **46** elevating the tongues **44**, and as a result, elevating the blower **18**, from the other portions of the external wall structure **14**. In one embodiment, the fan housing **12** is constructed of sheet metal and the pedestals **46** and tongues **44** are formed from the sheet metal of the associated external wall structure **14**. In this embodiment, the tongues **44** are stamped from the fan housing external wall structure **14** leaving an aperture in the fan housing external wall structure **14** adjacent to the tongue **44**.

[0028] In one embodiment, the tongues **44** extend from the fan housing external wall structure **14** at a tongue proximal end **52** and continue to a tongue distal end **54** at which the tongue **44** terminates, as depicted, for example, in FIGS. **3**C and **3**D. The depicted tongues **44** define a curvature creating a downward curve between the proximal end **52** and the distal end **54** to create a downward force on objects slide between the tongue and the fan housing external wall structure **14**.

[0029] To secure the blower **18** to the fan housing **12**, the blower **18** is first inserted into the fan housing **12** using a first motion in a first direction toward the fan housing **12** until each of the slots **42** are located around an associated one of the tongues **44**, as depicted in FIG. **3**C. This can be accomplished by inserting the blower **18** into the fan housing **12** in the first direction until the ears **40** contact the fan housing external wall structure **14** (e.g. the pedestals **46** in the depicted embodiment). Then, the blower **18** is moved using a second motion in a second direction, generally perpendicular to the first direction, toward the tongue proximal ends **52** until the scroll mounting plate **36** is shifted so that the slots **42** are slide out from under the tongues **44** and the tongues extend over a solid portion of the scroll mounting plate **36**, as depicted in FIG. **3**D, as well as FIG. **2**A. This can be accomplished by moving the blower **18** in the second direction until the edge of the slot **42** engages the tongue proximal end **52** and can travel no further. As a result of this second motion, the blower **18** is secured to the fan housing **12** and the force created by the curvature of the tongues **44** against the scroll mounting plate **36** keeps the blower **18** in place.

[0030] In this configuration, the blower **18** defines mounting holes **48** in either the scroll **30** or the scroll mounting plate **36** that are aligned with mounting holes **50** in the fan housing **12**. Fasteners such as screws or rivets can then be inserted through one or more of the mounting holes **48**, **50** more securely lock the blower in place relative to the fan housing **12**.

[0031] Although the tongues **44** described and depicted in the embodiments disclosed herein are one possible fastener facilitating the securement of the blower **18** to the fan housing **12**, alternative embodiments will be recognized by persons of ordinary skill in the art to allow installation of the blower **18** with the two-motion process described herein. In one non-limiting example, the tongues can be replaced by post extending from the fan housing **12** with an annular groove around the post configured to receive the scroll mounting plate after the slot is placed over the post. In this embodiment, the scroll mounting plate slots can optionally define keyhole shaped slots to accommodate the posts. Also, when tongues are used to facilitate this two-motion process, tongue configurations other than those shown and describe are also contemplated.

[0032] As depicted best in FIGS. **1** and **4**A-D, the blower outlet seat **24** is located between the scroll outlet **34** and the fan housing **12**. The blower outlet seat **24** is comprised of a plate **60** defining an external periphery **62** and an internal periphery **64**. The internal periphery **64** defines a blower outlet seat aperture **66**. The shape of the blower outlet seat aperture **66** complements the shape of the scroll outlet **34**. The internal periphery **64** of the blower outlet seat **24** defines a bevel **68** across a thickness of the plate **60**. Optionally, as depicted, a sleeve **70** can extend the beveled internal periphery **64** beyond the plate **60** on an exterior side **60***a* of the plate **60** to extend through the outlet aperture **22** of the fan housing **12**. The bevel **68** can extend beyond the internal periphery **64** and along all or a portion of the sleeve **70**. The size of the plate **60** can be used to cover a fan housing outlet **22** that is larger than the scroll outlet **34** such that different scrolls having different sized and shaped outlets can be accommodated into a single fan housing **12**.

[0033] The scroll outlet **34** is defined by a peripheral wall **80**. The peripheral wall **80** is shown as completely enclosing the scroll outlet **34**, but in alternative embodiments, the peripheral wall **80** could only partially enclose the scroll outlet **34**. At the scroll outlet **34**, the scroll peripheral wall **80** defines a bevel **82**. In one embodiment, the scroll peripheral wall bevel **82** complements the bevel **68** of the blower outlet seat **24**, as best depicted in FIG. **4**D. When the scroll **30** is installed in the fan housing **12** and the scroll peripheral wall bevel **82** engages the blower outlet seat bevel **68**, as depicted in FIG. **4**D, the scroll outlet **34** forms a seal with the blower outlet seat **24** and reduces or prevents air loss between the blower **18** and the fan housing **12**. While the scroll peripheral wall bevel **82** and the blower outlet seat bevel **68** are depicted as straight, other complementary configurations are also contemplated with the goal of creating a seal between the two bevels **68**, **82**. The angle of the bevels **68**, **82** can vary from the angle depicted and can also vary around the blower outlet seat aperture **68** and around the peripheral wall **80** of the scroll outlet **34**.

[0034] In one embodiment, the duct connector **26** of FIG. **4**A comprises a first connecting feature **100** depicted as a tongue. The first connecting feature **100** is located adjacent to the inlet aperture **90** of the duct connector **26**. A second connecting feature **100** (hidden by the blower outlet seat **24**) is located on the opposing side of the duct connector inlet aperture **90**. The first and second connecting features **100** can engage with a first shoulder **72** and a second shoulder **72** extending outward from the blower outlet seat **24** to connect the duct connector **26** to the blower outlet seat **24** and to locate the duct connector **26** with respect to the blower outlet seat **24** for optimal air flow. The first and second connecting features **100** can comprises elements other than the depicted tongue **100**.

[0035] In the depicted configuration in which the connecting features **100** are comprised of the tongue **100**, each tongue **100** can be slide over the shoulder **72** of the blower outlet seat **24** such that the shoulder **72** is slide between the tongue **100** and the remainder of the duct connector **26**. When the tongue **100** is slide over the shoulder **72** as far as it can go, the internal periphery **64** is aligned with the duct connector inlet aperture **90** for optimal flow. In one embodiment, once the duct connector **26** and the blower outlet seat **24** are connected and aligned in this manner, the duct connector **26** can be slide through the fan housing outlet **22** from the inside out until it abuts the fan housing and cannot proceed further. Connectors (e.g. screws, rivets, etc.) can then secure the blower outlet seat **24** and the duct connector **26** to the fan housing **12**, pinching the duct connector **26** between the blower outlet seat **24** and the fan housing **12**.

[0036] It will be understood that because the scroll peripheral wall bevel **82** extends into the blower outlet seat aperture **66**, as depicted in FIG. **4**D, the blower **18** cannot be installed with a single motion of lowering the blower **18** into the fan housing **12**, as is done with some existing blowers. Instead, installation requires a first motion lowering the blower **18** into the fan housing **12** in a first direction, and then, in a second motion, moving the blower **18** in a second direction, laterally, such that the scroll outlet **34** moves into the blower outlet seat aperture **66** to engage the bevels **68**, **82**.

[0037] It will also be understood that the first motion of securing the blower **18** to the fan housing is the same as the first motion of engaging the scroll peripheral wall bevel **82** and the blower outlet seat bevel **68**. Stated differently, by lowering the blower **18** into the fan housing **12**, two different things are accomplished: (i) the first step of locating the blower slots **42** over tongues **44** of the fan housing **12**, and (ii) the first step of engaging the scroll peripheral wall bevel **82** to the blower outlet seat bevel **68**. Additionally, the second motion of securing the blower **18** to the fan housing **12** (i.e. moving the blower sideways) accomplishes both sliding the ears **40** under the tongues **44**, and engaging the scroll peripheral wall bevel **82** with the blower outlet seat bevel **68**.

[0038] The scroll **30** can be made of any know materials. The blower outlet seat **24** can likewise be made of any known materials. The scroll **30** and the blower outlet seat **24** can be made of the same materials or of different materials. In one embodiment, the scroll **30** and the blower outlet seat **24** are both made of polypropylene.

[0039] The depicted duct connector **26** is one of any of the typical, known duct connectors and defines an inlet aperture **90**, with a flange **92** extending outwardly therefrom. The flange **92** engages with the fan housing **12** and facilitates securing the duct connector **26** to the fan housing **12**. A transition region **94** extends from the flange **90** and transitions the duct connector **26** from a rectangular shape to a round shape. A round region **96** extends from the round end of the transition region **94** and forms a round outlet aperture **98**. The external configuration of the blower outlet seat sleeve **70** preferably, complements the configuration of the inside of the duct connector inlet aperture **90** so that the two fit snuggly and create a seal there between.

[0040] In an alternative embodiment, not depicted, the blower outlet seat **24** may be integrally formed with the duct connector **26**. Integrating these two elements would reduce the number of components of the ventilation fan **10**.

[0041] The various components and features described above can be combined in a variety of ways, to provide other non-illustrated embodiments within the scope of the disclosure. As such, it is to be understood that the disclosure is not limited in its application to the details of construction and parts illustrated in the accompanying drawings and described hereinabove. The disclosure is capable of other embodiments and of being practiced in various ways. It is also to be understood that the phraseology or terminology used herein is for the purpose of description and not limitation.

[0042] Although the present disclosure has been described in the foregoing description by way of illustrative embodiments thereof, these embodiments can be modified at will, without departing from the spirit, scope, and nature of the subject disclosed.

[0043] Headings and subheadings, if any, are used for convenience only and do not limit the invention. The word exemplary is used to mean serving as an example or illustration. To the extent that the term include, have, or the like is used, such term is intended to be inclusive in a manner similar to the term comprise as comprise is interpreted when employed as a transitional word in a claim. Relational terms such as first and second and the like may be used to distinguish one entity or action from another without necessarily requiring or implying any actual such relationship or order between such entities or actions.

[0044] Phrases such as an aspect, the aspect, another aspect, some aspects, one or more aspects, an implementation, the implementation, another implementation, some implementations, one or more implementations, an embodiment, the embodiment, another embodiment, some embodiments, one or more embodiments, a configuration, the configuration, another configuration, some configurations, one or more configurations, the subject technology, the disclosure, the present disclosure, other variations thereof and alike are for convenience and do not imply that a disclosure relating to such phrase(s) is essential to the subject technology or that such disclosure applies to all configurations of the subject technology. A disclosure relating to such phrase(s) may apply to all configurations, or one or more configurations. A disclosure relating to such phrase (s) may provide one or more examples. A phrase such as an aspect or some aspects may refer to one or more aspects and vice versa, and this applies similarly to other foregoing phrases.

[0045] All numbers and ranges disclosed above may vary by some amount. Whenever a numerical range with a lower limit and an upper limit is disclosed, any number and any included range falling within the range are specifically disclosed. In particular, every range of values (of the form, "from about a to about b," or, equivalently, "from approximately a to b," or, equivalently, "from approximately a-b") disclosed herein is to be understood to set forth every number and range encompassed within the broader range of values. In addition, the terms in the claims have their plain, ordinary meaning unless otherwise explicitly and clearly defined by the patentee. Moreover, the indefinite articles "a" or "an," as used in the claims, are defined herein to mean one or more than one of the element that it introduces. If there is any conflict in the usages of a word or term in this specification and one or more patent or other documents that may be incorporated herein by reference, the definitions that are consistent with this specification should be adopted.

[0046] A phrase "at least one of" preceding a series of items, with the terms "and" or "or" to separate any of the items, modifies the list as a whole, rather than each member of the list. The phrase "at least one of" does not require selection of at least one item; rather, the phrase allows a meaning that includes at least one of any one of the items, and/or at least one of any combination of the items, and/or at least one of each of the items. By way of example, each of the phrases "at least one of A, B, and C" or "at least one of A, B, or C" refers to only A, only B, or only C; any combination of A, B, and C; and/or at least one of each of A, B, and C.

[0047] In one aspect, a term coupled or the like may refer to being directly coupled. In another aspect, a term coupled or the like may refer to being indirectly coupled. Terms such as top, bottom, front, rear, side, horizontal, vertical, and the like refer to an arbitrary frame of reference, rather than to the ordinary gravitational frame of reference. Thus, such a term may extend upwardly, downwardly, diagonally, or horizontally in a gravitational frame of reference.

[0048] The title, background, brief description of the drawings, abstract, and drawings are hereby incorporated into the disclosure and are provided as illustrative examples of the disclosure, not as restrictive descriptions. It is submitted with the understanding that they will not be used to limit the scope or meaning of the claims. In addition, in the detailed description, it can be seen that the description provides illustrative examples and the various features are grouped together in various implementations for the purpose of streamlining the disclosure. The method of disclosure is not to be interpreted as reflecting an intention that the claimed subject matter requires more features than are expressly recited in each claim. Rather, as the claims reflect, inventive subject matter lies in less than all features of a single disclosed configuration or operation. The claims are hereby incorporated into the detailed description, with each claim standing on its own as a separately claimed subject matter.

[0049] The use of the terms "a" and "an" and "the" and "said" and similar references in the context of describing the invention (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. An element proceeded by "a," "an," "the," or "said" does not, without further constraints, preclude the existence of additional same elements. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., "such as") provided herein, is intended merely to better illuminate the disclosure and does not pose a limitation on the scope of the disclosure unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the disclosure.

[0050] Numerous modifications to the present disclosure will be apparent to those skilled in the art in view of the foregoing description. Preferred embodiments of this disclosure are described herein, including the best mode known to the inventors for carrying out the disclosure. It should be understood that the illustrated embodiments are exemplary only, and should not be taken as limiting the scope of the disclosure.

We claim:

1. A ventilation fan configured to be positioned adjacent to a room of a building structure to ventilate the room, the ventilation fan comprising:
- a fan housing comprising an external wall defining an internal region;
- a blower configured to be located in the internal region;
- a tongue extending from one of the fan housing and the blower;
- a slot defined by one of the fan housing and the blower, the slot configured to receive the tongue to secure the blower to the fan housing.

2. The ventilation fan of claim **1**, wherein the blower comprises a scroll and a scroll mounting plate, and the slot is defined in the scroll mounting plate.

3. The ventilation fan of claim **1**, further comprising a second tongue and a third tongue.

4. The ventilation fan of claim **1**, further defining a second slot and a third slot.

5. The ventilation fan of claim **2**, wherein the tongue comprises a proximal end at the fan housing and a distal end spaced from the fan housing, and the scroll mounting plate sits between the tongue distal end and the external wall when the blower is secured to the fan housing.

6. The ventilation fan of claim **2**, wherein the scroll mounting plate comprises an ear extending beyond the perimeter of the scroll and the slot is defined in the ear.

7. The ventilation fan of claim **2**, wherein the blower is secured to the fan housing by sliding the scroll mounting plate under the tongue.

8. A ventilation fan configured to be positioned adjacent to a room of a building structure to ventilate the room, the ventilation fan comprising:
- a fan housing comprising an external wall defining an internal region, a fan housing inlet to the internal region and a fan housing outlet from the internal region;
- a blower configured to be located in the internal region, the blower comprising a scroll defining a blower inlet through which the blower is configured to draw air and a blower outlet through which the blower is configured to expel air, the blower outlet defining a blower outlet bevel; and
- an outlet seat between the blower outlet and the fan housing outlet, the outlet seat defining an aperture complementing the shape of the blower outlet and defining an outlet seat bevel;
- wherein the blower outlet bevel is configured to engage the outlet seat bevel.

9. The ventilation fan of claim **8**, wherein the outlet seat bevel complements the blower outlet bevel.

10. The ventilation fan of claim **8**, further comprising a duct connector that comprises the outlet seat.

11. The ventilation fan of claim **8**, wherein the outlet seat comprises a sleeve configured to extend into the fan housing outlet and the outlet seat bevel extends into the sleeve.

12. The ventilation fan of claim **8**, wherein the outlet seat comprises a plate defining the outlet seat aperture and a sleeve extending from the plate at the outlet seat aperture, wherein the sleeve comprises at least part of the outlet seat bevel, and the outlet seat bevel complements the blower outlet bevel.

13. The ventilation fan of claim **12**, the plate of the outlet seat is secured to the fan housing.

14. The ventilation fan of claim **8**, further comprising a duct connector configured to be secured to the exterior of the fan housing, wherein the outlet seat comprises a sleeve configured to extend into the fan housing outlet and contact the duct connector outside of the fan housing.

15. A method for assembling a ventilation fan having a fan housing defining an internal region, and a blower, the method comprising the steps of:
- (a) moving the blower in a first direction toward the fan housing to place the blower into the internal region;
- (b) moving the blower a second direction, generally perpendicular to the first direction to secure the blower to the housing.

16. The method of claim **15**, wherein the fan housing comprises a tongue and the blower comprises a slot and step (a) further comprises locating the slot around the tongue.

17. The method of claim **16**, wherein the tongue extends from the fan housing into the internal region and step (a) further comprises moving the blower in the first direction until the blower contacts the fan housing.

18. The method of claim **16**, wherein step (b) further comprises moving the blower in the second direction until the blower is under the tongue.

19. The method of claim **16**, wherein the blower comprises an ear in which the slot is defined and wherein step (b) further comprises moving the blower in the second direction until the ear is between the tongue and the fan housing.

20. The method of claim **19**, wherein the tongue extends from the fan housing at a tongue proximal end and extends to a tongue distal end and wherein step (b) further comprises moving the blower in the second direction until the ear contacts the tongue proximal end.

* * * * *